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Distributions and sampling: the Gaussian, the central limit theorem, A/B tests and drawing the next word

Why can the same question get a different answer from a chat model, and how do we test whether a new AI feature is better?

Mathematics for AI and Machine Learning | Session 14 of 18 | Probability and statistics

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Today in one view: what you will leave with

By the end of today you can



Describe the Bernoulli, binomial and Gaussian distributions, and work a binomial probability with factorials

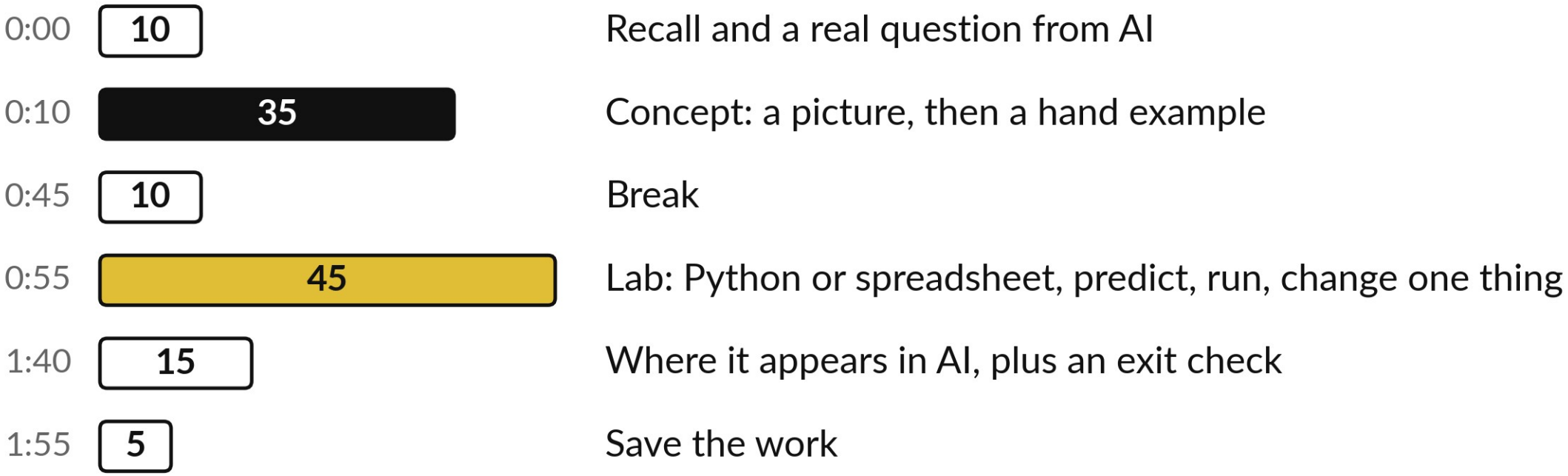


Explain the law of large numbers, the central limit theorem, standard error and a 95% confidence interval, and run a two-proportion A/B test



Draw the next word from a probability list with a random number, and explain why the same question can get different answers

Your two hours: distributions, averages, a test, then drawing a word



Pictures and hand calculations first. Then the lab on averages, an A/B test and drawing words.

Three questions from last time

1. Two fair dice are independent. What does that mean, and what is P(both show 6)?

2. What is the expectation of a fair die, and what does it mean?

3. For six learners' scores, do we divide the variance by n or by n - 1, and why?

Answer aloud first. Then we check together.

Answers to the three questions from Session 13

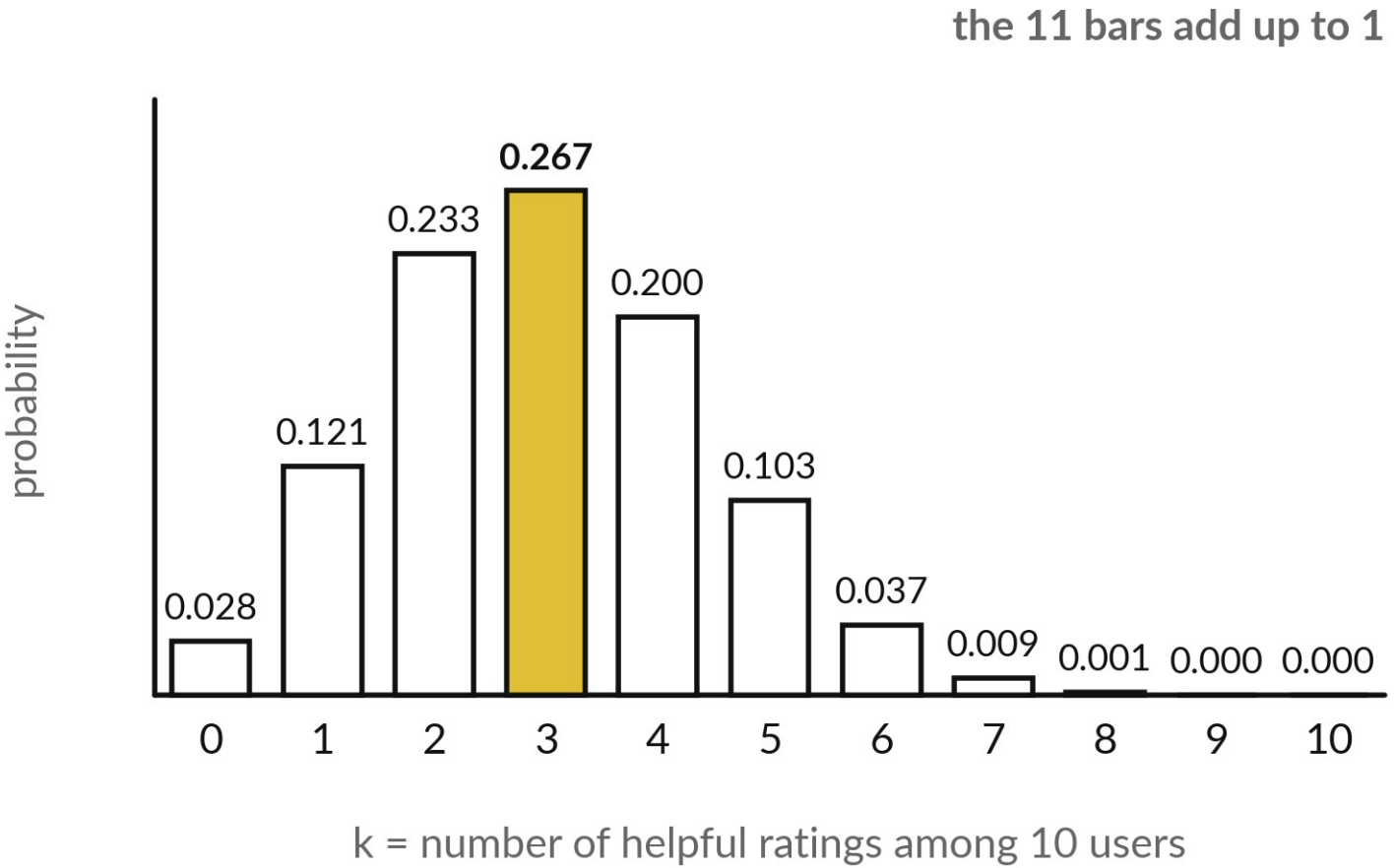
Question	Short answer
Two fair dice are independent. What does that mean, and what is P(both show 6)?	Knowing one die does not change the probabilities of the other. P = 1/6 × 1/6 = 1/36.
What is the expectation of a fair die, and what does it mean?	3.5. It is the probability-weighted average over many rolls. No single roll shows it.
For six learners' scores, do we divide the variance by n or by n - 1, and why?	By n - 1, because the six are a sample (VAR.S, ddof=1). Dividing by n treats the list as the whole population.

If an answer surprised you, look back at Session 13 before the lab.

Words we use today

	Word	What it means
	**Binomial distribution**	The number of successes in n independent yes or no trials, each with probability p.
	**Gaussian (normal) distribution**	The bell curve, set by its mean mu and standard deviation sigma.
	**Central limit theorem**	Averages of many independent values are close to Gaussian, whatever the original shape.
	**Standard error, confidence interval, p-value**	How much a sample mean wanders, a range built from it, and how surprising a gap would be if nothing changed.
	**Sampling**	Drawing a value at random according to its probabilities, as a chat model draws its next word.

A Bernoulli trial is one yes or no, and a binomial counts the yeses in n independent trials



Illustrative: a user rates an answer helpful with p = 0.3. X counts the helpful ratings of 10 users.

One user: Bernoulli(p). Mean p, variance p(1 - p) = 0.21.

Independent (Session 13): one rating does not change another.

P(X = 3) = 0.2668 (gold). Mean n p = 3, sd 1.449.

The binomial formula counts the ways with factorials, then multiplies the probabilities

Binomial pmf

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

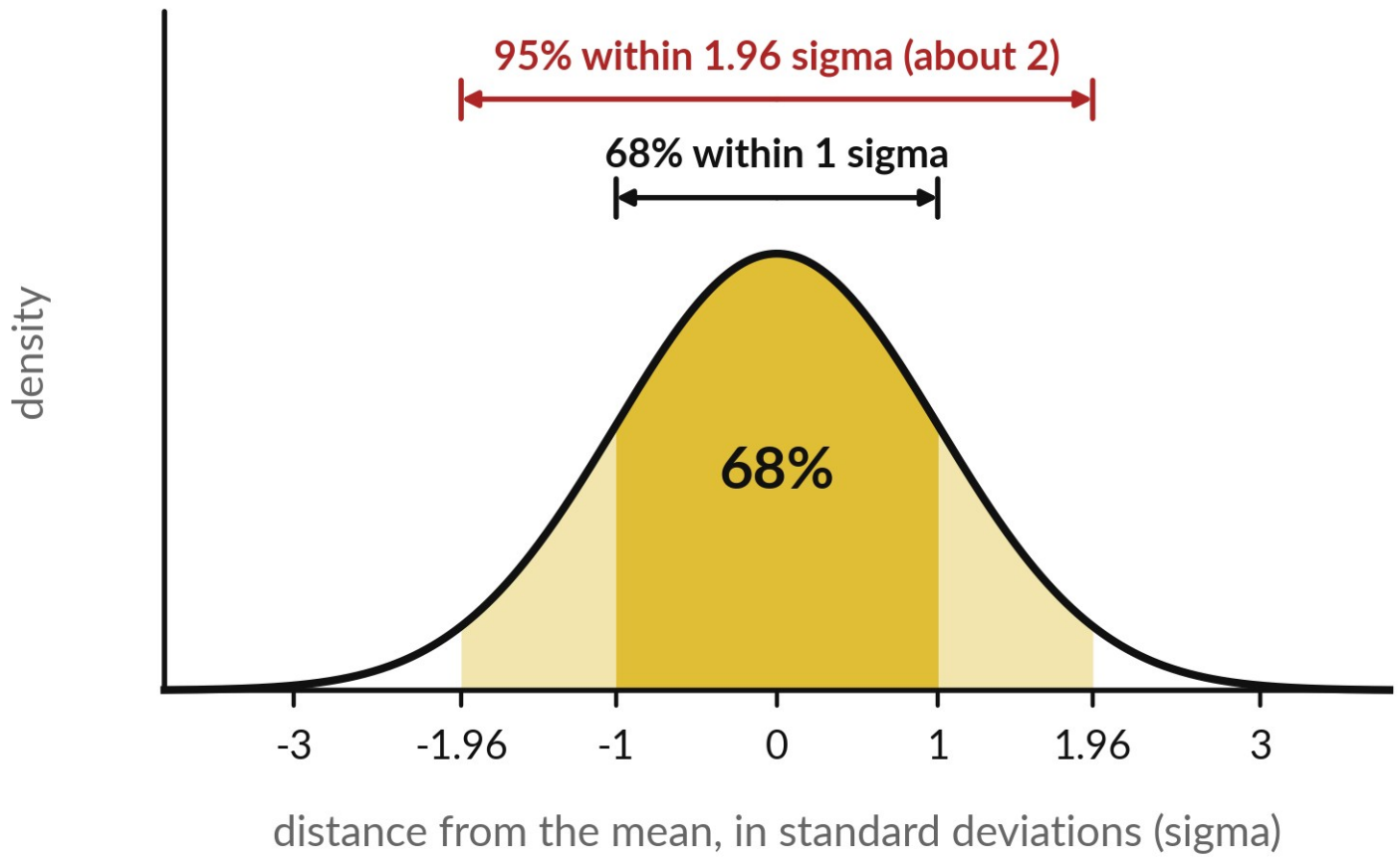
The ways, by hand

$$\binom{10}{3} = \frac{10!}{3!\ 7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = \frac{720}{6} = 120$$

Symbol	What it means	In our data
n! (n factorial)	Multiply every whole number from n down to 1. 0! = 1.	3! = 3 × 2 × 1 = 6
n choose k	The number of ways to place k successes among n trials: n! / (k! (n - k)!).	10 choose 3 = 120
p to the k, (1-p) to the n-k	The k trials all succeed, and the other n - k all fail.	0.027 and 0.08235
P(X = 3)	Multiply the three parts.	120 × 0.027 × 0.08235 = 0.2668

Spreadsheet: COMBIN(10, 3) = 120. BINOM.DIST(3, 10, 0.3, FALSE) gives the whole answer. TRUE gives P(X at most 3).

The Gaussian (normal) distribution is the bell curve, and 95% of it lies within 1.96 standard deviations



Gaussian (normal): a symmetric bell, centred on the mean mu, width set by sigma.

About 68% lies within 1 sigma of mu.

95% lies within 1.96 sigma. Every 95% interval today uses this 1.96.

Averages of many small effects look Gaussian.

The Gaussian density, the z-score and the area to the left of z

Gaussian pdf, z-score

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad z = \frac{x-\mu}{\sigma}$$

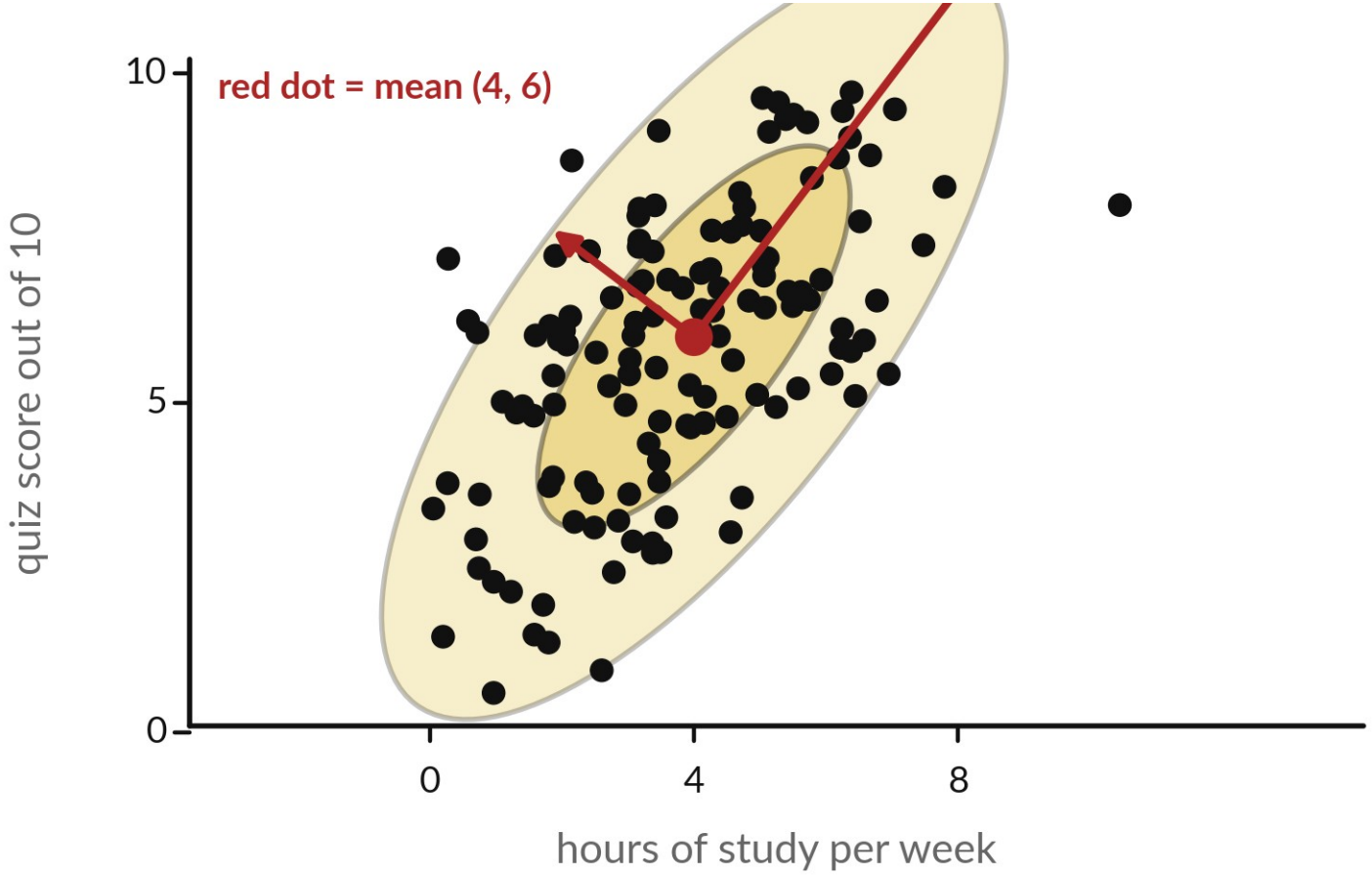
Area left of z

$$\Phi(z) = P(Z \le z) = \frac{1}{2}\left(1 + \operatorname{erf}\left(z/\sqrt{2}\right)\right)$$

Symbol	What it means	In our data
mu, sigma	Quiz scores (illustrative): Normal with mean 60 and standard deviation 10.	mu = 60, sigma = 10
z	How many standard deviations x lies above the mean.	x = 75 gives z = 1.5
Phi(z)	The area to the left of z. erf is a built-in function that computes it.	Phi(1.5) = 0.9332

Areas: between z = −1 and 1, 0.6827; between −1.96 and 1.96, 0.9500. Sheet: NORM.DIST(75, 60, 10, TRUE) = 0.9332.

Recall: two columns at once form a tilted Gaussian cloud, described by the covariance matrix



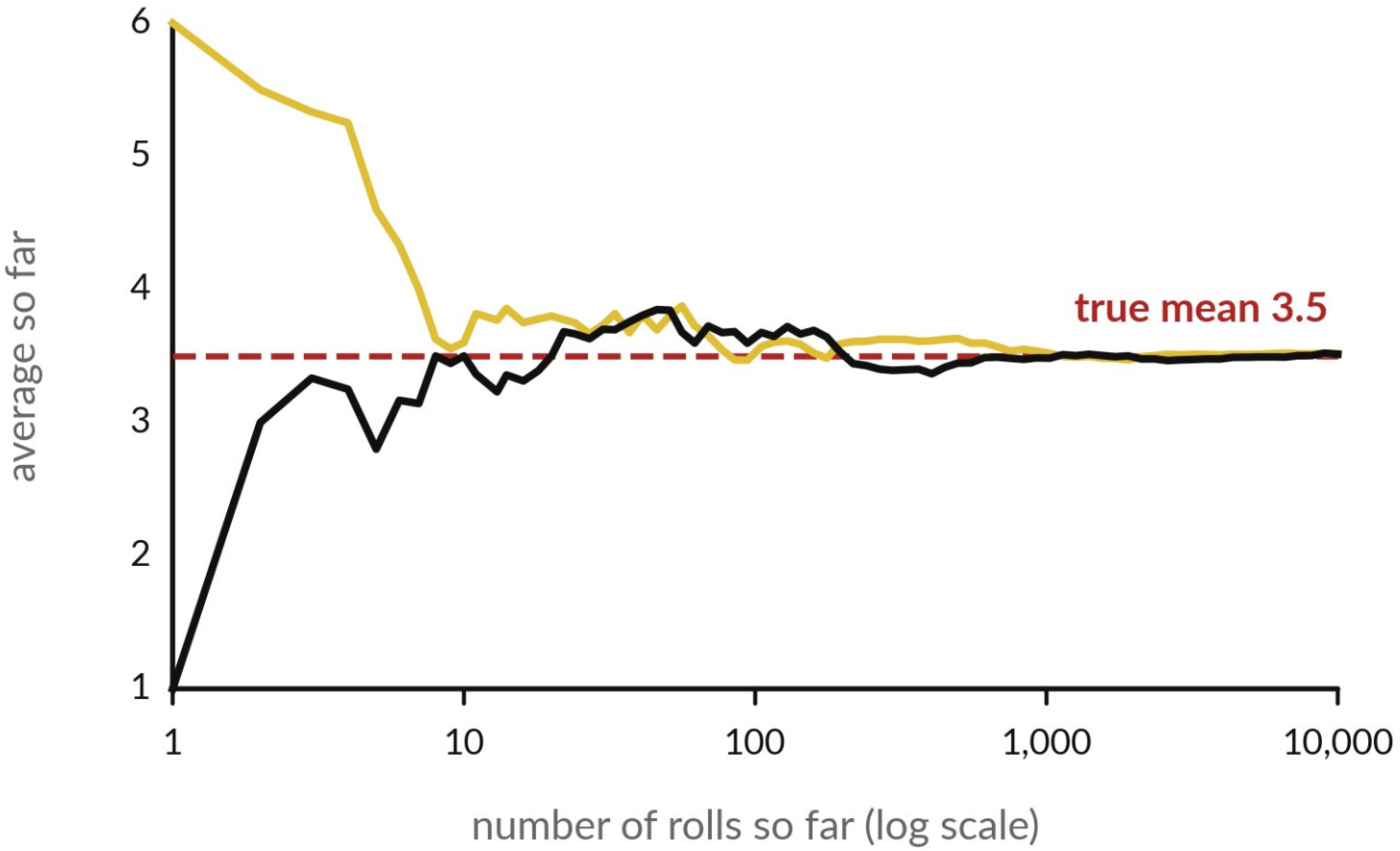
Means (4, 6). Sample covariance matrix (Session 13): variances 5.6 and 8.4, covariance 5.

Ellipses: 1 and 2 sd. Red arrows: eigenvectors (PCA, Session 6).

Eigenvalues 12.19 and 1.81: the variance along each arrow.

130 of 150 simulated points: scores outside 0 to 10 are left out.

The law of large numbers: the more repeats, the closer the average gets to the true mean



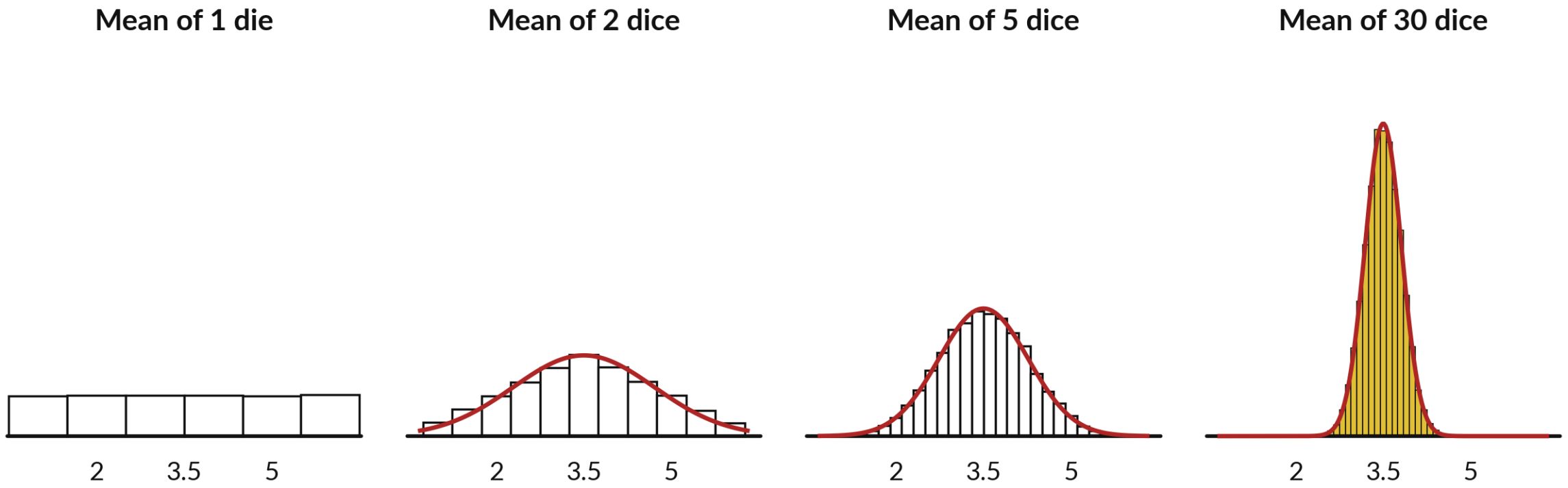
Two runs of rolling a die 10,000 times (simulation). The line is the running average.

After 1 roll the average is one of 1 to 6, often far from 3.5.

After 100 rolls it is near 3.5.

After 10,000 rolls it is usually within about 0.03 of 3.5.

The central limit theorem: averages of many rolls look Gaussian, though a single die is flat



Each panel: 20,000 averages of n dice. n = 1 is flat, not Gaussian. At n = 30 the red curve fits.

Standard error: how much a sample mean wanders

Central limit theorem

$$\bar{X} \approx \text{Gaussian with mean } \mu \text{ and standard deviation } \sigma/\sqrt{n}$$

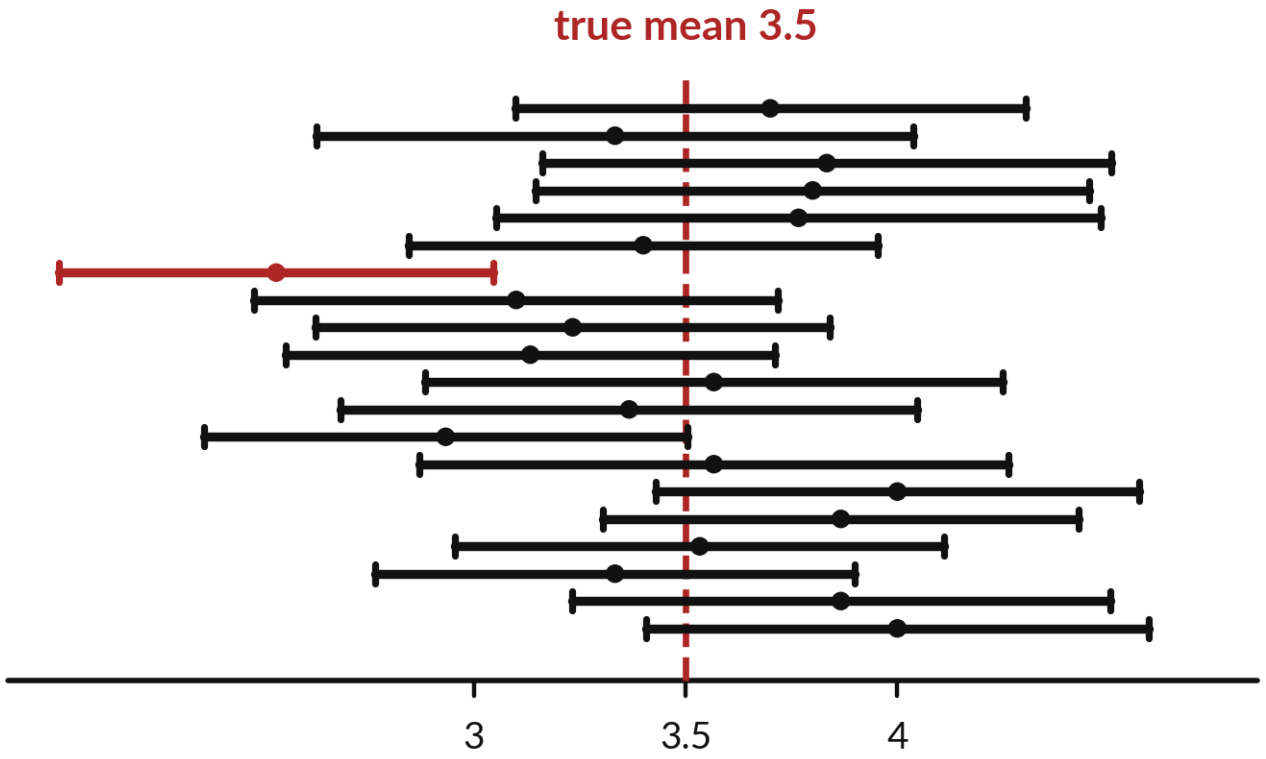
Standard error

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

Symbol	What it means	In our data
x-bar	The mean of a sample of n independent values.	mean of 30 dice
sigma, n	Standard deviation of one value, and the sample size. The die's sigma is known (Session 13), so no n - 1 here.	sigma = 1.7078
SE	Standard error: the standard deviation of the sample mean.	1.7078 / sqrt(30) = 0.3118

Four times the data halves the SE. Simulated 30-dice averages (seed 14): standard deviation 0.311.

A 95% confidence interval comes from a method that catches the true mean in about 95 of 100 samples



mean of 30 rolls, with its 95% interval

20 samples of 30 dice (simulation). 19 of the 20 intervals contain the true mean 3.5. The red one misses.

Interval = sample mean plus or minus 1.96 standard errors: the 95% band of the Gaussian.

The 95% describes the method over many samples.

It is not the chance that this one interval holds the true value.

Confidence interval for a mean, worked on the 60 fixed rolls

Interval $\bar{x} \pm 1.96\,\mathrm{SE}, \qquad \mathrm{SE} = \frac{s}{\sqrt{n}}$

Symbol	What it means	In our data
x-bar, n	The mean of the 60 rolls from Session 13, and the sample size.	3.9667, 60
s	Sample standard deviation: sigma is unknown, so estimate it with n - 1 (STDEV.S).	1.7847
SE	Standard error s over the square root of n.	0.2304
1.96	Phi(1.96) = 0.975, so 95% of a Gaussian lies between −1.96 and 1.96.	95% band
Interval	Mean plus or minus 1.96 SE.	3.515 to 4.418

With only 60 rolls a statistician uses the t value 2.00: 3.506 to 4.428, and 3.5 is still just outside.

An A/B test splits users at random into two groups and compares one rate

	A	B	C	D
	Group	**Users**	**Rated helpful**	**Rate**
1	A: current app	2,400	288	12.0%
2	B: new feature	2,400	336	14.0%

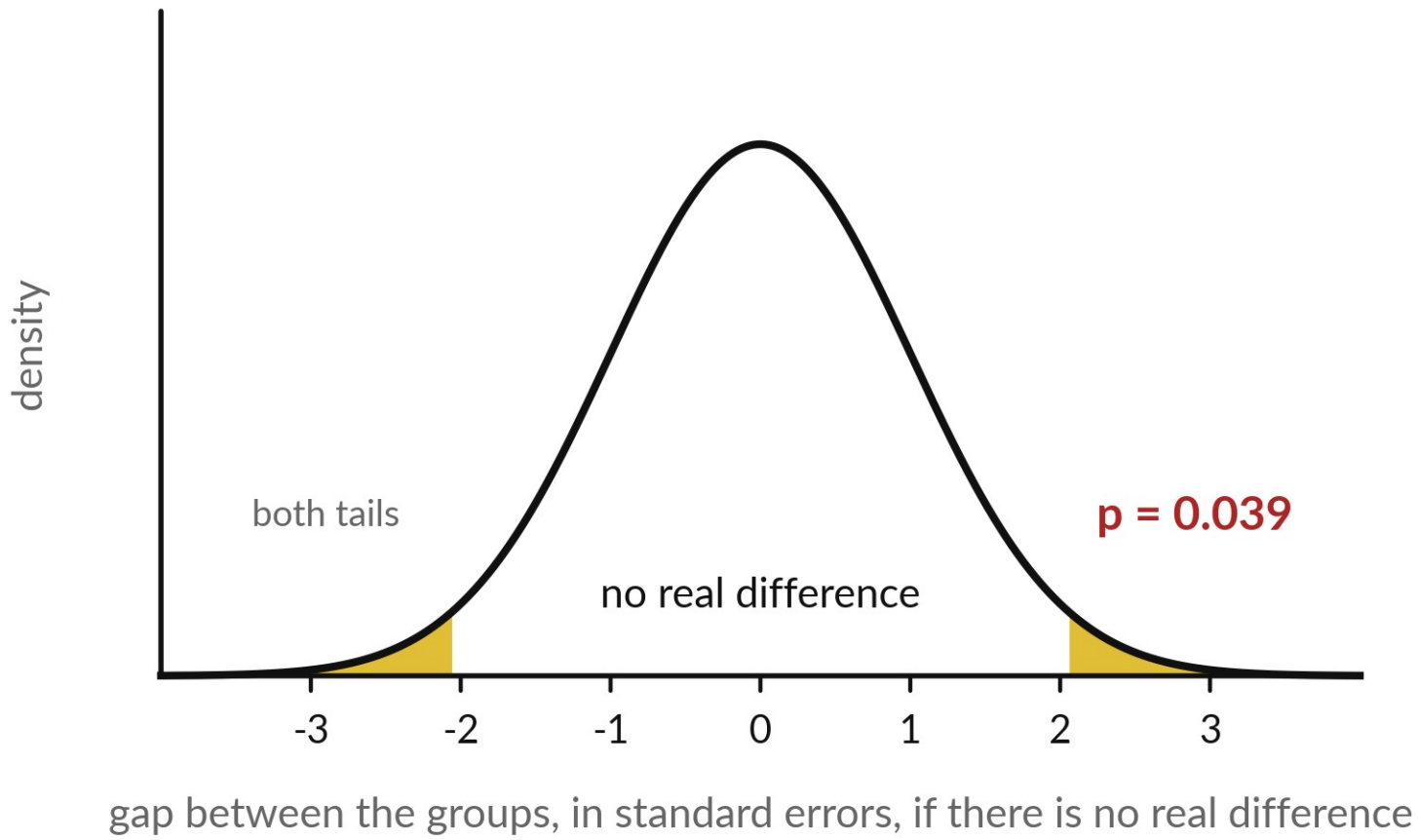
Illustrative counts: a new suggested follow-up feature in an AI chat app. Metric: share of chats rated helpful.

Random assignment makes the groups alike, apart from the feature.

B beats A by 2.0 percentage points.

Question: a real effect, or luck?

A hypothesis test asks how likely a gap this large would be if the feature did nothing



Null hypothesis: the true rates are equal. Our gap is z = 2.06 standard errors.

The gold tails hold gaps at least this large, in either direction. Their area is the p-value: 0.039.

A common, arbitrary convention calls $p < 0.05$ "significant".

Small p-value: this gap would be rare by luck alone.

The two-proportion z-test

Standard error under the null

$$\mathrm{SE} = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_A}+\frac{1}{n_B}\right)}$$

z statistic

$$z = \frac{\hat{p}_B - \hat{p}_A}{\mathrm{SE}}$$

Symbol	What it means	In our data
p-hat_A, p-hat_B	The rate in each group: helpful chats k divided by users n.	288 / 2400 = 0.12, 336 / 2400 = 0.14
p-hat_pool	The pooled rate: (k_A + k_B) / (n_A + n_B), both groups together.	0.130
SE and z	The standard error of the gap, and the gap in standard errors.	0.00971 and 2.060

Two-sided p-value = 2 × (1 - Phi(z)) = 0.0394. Spreadsheet: 2*(1-NORM.DIST(z,0,1,TRUE)).

The A/B test step by step, with a 95% interval for the gap

Step	Calculation	Result
1 Rates	288 / 2400 and 336 / 2400	0.120 and 0.140
2 Pooled rate	(288 + 336) / (2400 + 2400)	0.130
3 Standard error	square root of 0.13 × 0.87 × (1/2400 + 1/2400)	0.00971
4 z statistic	0.020 / 0.00971	2.060
5 p-value	2 × (1 - area to the left of z)	0.0394
6 Interval for the gap	0.020 plus or minus 1.96 × 0.00970	0.0010 to 0.0390

Step 3 pools because the test assumes no difference. Step 6 does not assume that: SE = sqrt(p_A(1-p_A)/n_A + p_B(1-p_B)/n_B). The interval, 0.1 to 3.9 points, only just excludes 0.

A p-value is easy to misread

A p-value is

The probability of a gap at least this large, either way, if there were truly no difference

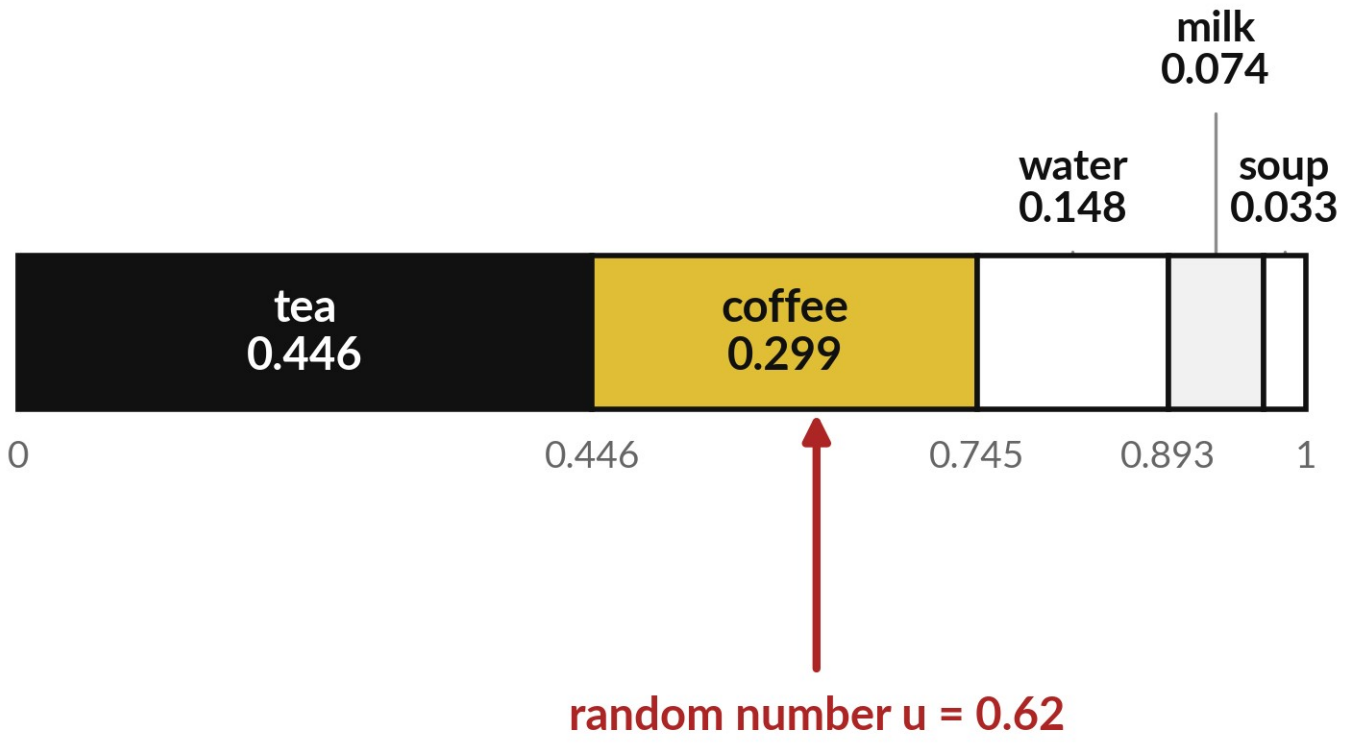
A p-value is not

The probability that the feature does nothing

The size of the effect

Testing many features: of 20 useless ones, about 1 passes the 0.05 line by luck. Report the gap and its interval.

Sampling from a distribution: cut 0 to 1 into slices and see where a random number lands



The next-word list of Session 13. Each word owns a slice as wide as its probability. Draw a random number u between 0 and 1.

u = 0.62 lands in the coffee slice (0.446 to 0.745).

Draw again and u differs, so the word can differ.

Likely words are picked often, unlikely ones sometimes.

Bridge: sample the central limit theorem and the next word with NumPy

```
rng = np.random.default_rng(14)
means = rng.integers(1, 7, (20000, 30)).mean(axis=1)
means.mean(), means.std(ddof=1)
np.cumsum([1, 2, 3])      # running totals: 1, 3, 6
p = np.array([0.446, 0.299, 0.148, 0.074, 0.033])
cum = np.cumsum(p)        # slice edges
np.searchsorted(cum, 0.62)
```

In plain words

Line 2: 20,000 samples of 30 dice. mean(axis=1) averages each row.

Line 3: the mean is near 3.5; the standard deviation is near the standard error 0.3118 (0.311 with seed 14).

Line 4: np.cumsum gives running totals.

Lines 6 and 7: the slice edges, then the first edge that reaches u: 1 = coffee.

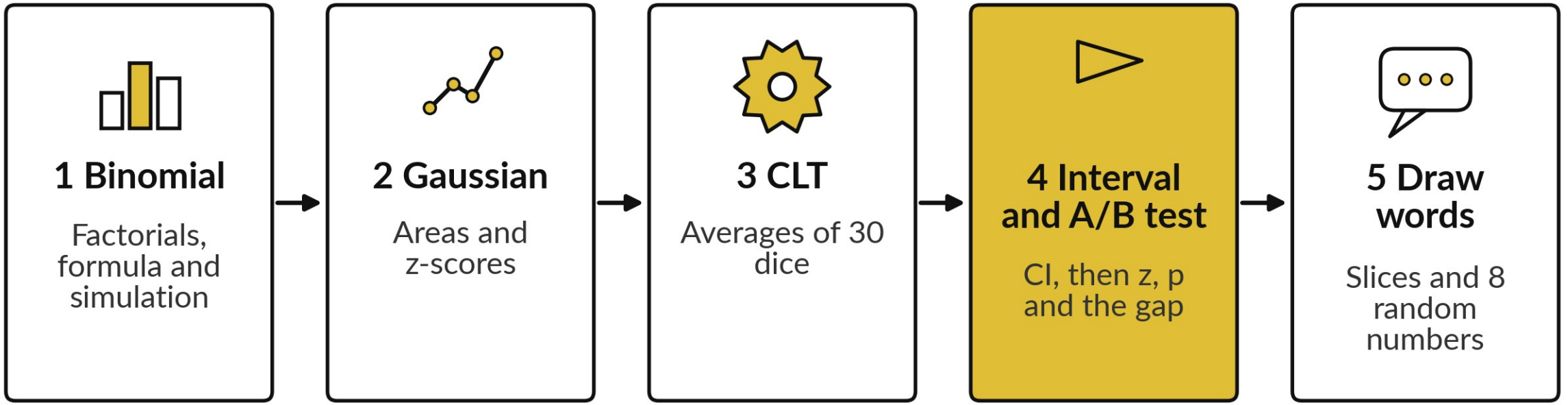
Spreadsheet: NORM.DIST, BINOM.DIST and fixed sample columns replace the random draws.

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Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has five stages, in Python and in the spreadsheet



Python notebook: Distributions and sampling. Workbook sheets: Binomial, Normal, Sampling, ABtest and Words. Predict, run, read, change one thing.

Check your numbers against these before you move on

What you compute	You should see
Binomial(10, 0.3): 10 choose 3, and P(X = 3)	120 and 0.2668
Normal (mean 60, sd 10): P(X below 75), P(X from 50 to 70)	0.9332 and 0.6827
Standard error of the mean of 30 dice: formula, and simulated (seed 14)	0.3118; about 0.311
Confidence interval of the 60 fixed rolls (s with n - 1)	3.515 to 4.418
A/B test: pooled rate, z, two-sided p-value	0.13, 2.060, 0.0394
A/B test: 95% interval for the gap	0.0010 to 0.0390
Words: running total after coffee, and tea in the 8 draws	0.7448 and 4

Simulated numbers use the seeds in the notebook. They match the slides only if the seed line runs first.

Errors you will meet, what they mean and what to do

	Where	What you see	What to do
✗	**Python**	Your sample numbers differ from the slides	Use Run all from the top. Every random step makes its own seeded generator.
✗	**Python**	p-value 0.0197 instead of 0.0394	You used one tail. Double the tail area for a two-sided test.
✗	**Spreadsheet**	BINOM.DIST gives P(X at most 3) = 0.6496	The last argument was TRUE. Use FALSE for exactly 3.
✗	**Both**	Interval 3.519 to 4.415, slightly narrow	You divided by n. The sample s uses n - 1: STDEV.S or ddof=1.
✗	**Both**	A draw picks the word one slice too late	u belongs to the first slice whose running total reaches u.

Where this appears in AI: a language model draws its next token at random from its probabilities

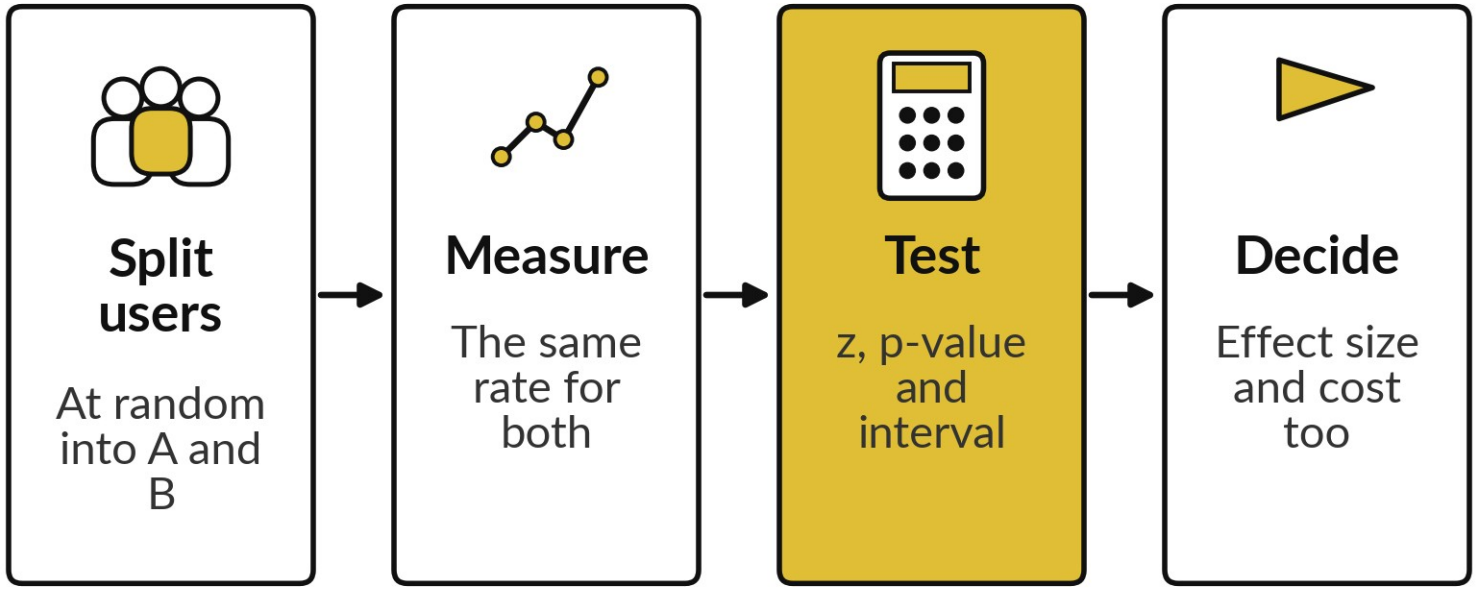


Published language models produce a probability for every token, and sampling draws one of them.

A new random number each time: the same question can get a different answer.

Public APIs document settings that reshape the list first: temperature and top-p (Session 17).

Where this appears in AI: an A/B test shows whether a new AI feature really is better



Online experiments compare two versions of a product on one measured rate.

Published practice: Kohavi, Tang and Xu describe it in their 2020 book on online controlled experiments.

The mathematics is the same as today. Which AI features a company tests is its own business.

Where this appears in AI: check an assistant with your numbers

8 minutes Check an AI assistant

Ask an AI assistant to run the A/B test on our made-up counts.

Give it A: 288 of 2,400 and B: 336 of 2,400.

Ask for z, p and a 95% interval.

Compare with 2.06, 0.039 and 0.1 to 3.9 points.

Ask what p means. Look for misreadings.

You keep

A note: its numbers, your numbers, and any misreading you spotted.

Exit check: three questions

Question	Short answer
A fair coin is flipped 10 times. What are the mean and standard deviation of the number of heads?	n p = 5 and the square root of 10 × 0.5 × 0.5, which is 1.581.
A test gives p = 0.03. Does that mean a 3% chance the feature does nothing?	No. It is the chance of a gap this large if there is truly no difference.
With the slices of today, which word does u = 0.90 pick, and why can the next answer differ?	Milk (0.893 to 0.967). The next draw uses a new random number.

Save the lab and keep your key numbers

What you keep from today

Keep your saved lab and these numbers: binomial P(X = 3) = 0.2668, normal P(X < 75) = 0.9332, the interval of the 60 rolls 3.515 to 4.418, A/B test z = 2.06 and p = 0.039, and tea in 4 of the 8 draws.

Our question today

Why can the same question get a different answer from a chat model, and how do we test whether a new AI feature is better?

Next session

Session 15: conditional probability and Bayes. We update beliefs with evidence and build a small spam filter by hand.

Take away

Saved lab
Notebook or workbook

Key numbers
See the check table

Next session
Bayes theorem

Bring
A question about a test result